

UCT701: USABILITY DESIGN OF SOFTWARE APPLICATIONS

L	T	P	Cr
2	0	2	3.0

Course Objectives: The major emphasis of the course will be on creating a learning system through which management students can enhance their innovation and creative thinking skills, acquaint themselves with the special challenges of starting new ventures and use IPR as an effective tool to protect their innovations and intangible assets from exploitation.

Introduction to User Centred Design

Aspects of User Centred Design: Product Appreciation Assignment – Evaluating the product from user centred design aspects such as functionality, ease of use, ergonomics, aesthetics.

Heuristic Evaluation: 10 Heuristic Principles, Examples Heuristic Evaluation: Group Assignment initiation (Website and App), Evaluation for key tasks of the app or website for heuristic principles, severity, recommendations.

Group Assignment Presentations and reviews

Group Project identification: Students will identify a project such as a website or mobile app to redesign. They will take this redesign project through the design lifecycle:

- Discovery
- Define
- Design
- Implement (Design Prototype)
- Usability Testing

The below design methods and techniques will be imparted w.r.t. the group project selected by the students.

UX Research: Understanding users, their goals, context of use, and environment of use.

Research Techniques: Contextual Enquiry, User Interviews, Competitive Analysis for UX.

Scenarios and Persona Technique

Presentation of Personas for the group project

Design Thinking Technique

Discovery and brainstorming

Concept Development

Task flow detailing for the Project

Prototyping Techniques: Paper, Electronic, Prototyping Tools

Project Prototyping Iteration 1

Project Prototyping Iteration 2

Review and feedback

Course Learning Outcomes (CLOs) / Course Objectives (COs):

Upon completion of this course, students will be able to:

1. To sensitise the students to the fundamentals of User Centred Design, its relevance and contribution to businesses.
2. To familiarize them with heuristic principles for effective design of software applications.
3. To understand the user research methods for a better user experience.
4. To acquire knowledge in redesigning an existing application or website.

Text Books:

1. Class Handouts
2. Online forum links, reference articles, blogs
3. Interaction Design: Beyond Human-Computer Interaction, 4th Edition, Jenny Preece, Helen Sharp and Yvonne Rogers
4. About Face, 4th Edition, Alan Cooper and Robert Reimann
5. Observing the User Experience, Second Edition: A Practitioner's Guide to User Research. Elizabeth Goodman, Mike Kuniavsky, Andrea Moed
6. The Elements of User Experience: User-Centered Design for the Web and Beyond. 2nd Edition, Jesse James Garrett
7. Understanding Design Thinking, Lean, and Agile - [Jonny Schneider](#)